CBSE Class 12 Pe — Half-Yearly Predicted Paper (Set 7 of 10)

SECTION A: Objective Type Questions (18 Marks: 18 x 1)

All questions are compulsory. Consists of MCQs and Assertion-Reason questions.

Q1. Which of the following is an example of a Combination Tournament? (A) Single League Tournament (B) Knock-out cum League (C) Consolation Tournament (D) Challenge Tournament *Unit 1: Management of Sporting Events | Tag: Very High***

Q2. Which postural deformity is caused by the severe deficiency of Vitamin D, Calcium, and Phosphorus, resulting in an outward bowing of the legs? (A) Knock Knees (B) Flat Foot (C) Bow Legs (Genu Varum) (D) Scoliosis *Unit 2: Children & Women in Sports | Tag: Very High***

Q3. Which of the following yoga asanas is strictly performed in a prone position (lying face down on the stomach)? (A) Halasana (B) Bhujangasana (C) Vajrasana (D) Tadasana *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | Tag: High***

Q4. Who among the following is the founder of the Special Olympics movement? (A) Baron Pierre de Coubertin (B) Sir Ludwig Guttmann (C) Eunice Kennedy Shriver (D) John Fontanella *Unit 4: Physical Education & Sports for CWSN | Tag: Very High***

Q5. Which essential macro-mineral is primarily responsible for maintaining fluid balance in the cells and transmitting nerve impulses during sports activities? (A) Iron (B) Sodium (C) Iodine (D) Copper *Unit 5: Sports & Nutrition | Tag: High***

Q6. The primary equipment required to accurately conduct the 'Sit and Reach Test' for flexibility is a: (A) Measuring tape and a cone (B) Stopwatch and a whistle (C) Specially constructed wooden testing box (D) Horizontal bar *Unit 6: Test and Measurement in Sports | Tag: Very High***

Q7. A bone injury where the bone fractures at an exact right angle (90°) to the long axis of the bone shaft is medically classified as a: (A) Transverse Fracture (B) Oblique Fracture (C) Greenstick Fracture (D) Comminuted Fracture *Unit 7: Physiology & Injuries in Sport | Tag: Very High***

Q8. The specific type of friction that occurs when an athlete moves through a liquid or a gas (e.g., a swimmer moving through water) is known as: (A) Static friction (B) Rolling friction (C) Sliding friction (D) Fluid friction *Unit 8: Biomechanics and Sports | Tag: Very High***

Q9. Which organizing committee is strictly responsible for using the public address system to inform athletes about upcoming events and track events during a sports meet? (A) Reception Committee (B) Announcement Committee (C) Transport Committee (D) Finance Committee *Unit 1: Management of Sporting Events | Tag: High***

Q10. A significant decrease in bone mineral density and bone mass, which makes female athletes highly susceptible to stress fractures, is known as: (A) Amenorrhea (B) Osteoporosis

(C) Anorexia Nervosa (D) Dysmenorrhea *Unit 2: Children & Women in Sports | Tag: Very High***

Q11. What is the primary focus of *Nadi Shodhana* in yoga practices? (A) It is an asana for muscle strength. (B) It is a Kriya for stomach cleansing. (C) It is a Pranayama (breathing technique) for purifying the energy channels. (D) It is a mudra for joint flexibility. *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | Tag: Very High***

Q12. Which of the following is NOT a recognized international sports organization for individuals with disabilities? (A) Deaflympics (B) Special Olympics (C) Paralympics (D) FIFA *Unit 4: Physical Education & Sports for CWSN | Tag: High***

Q13. Which specific vitamin is crucial for the coagulation (clotting) of blood, preventing excessive bleeding from sports injuries? (A) Vitamin A (B) Vitamin C (C) Vitamin E (D) Vitamin K *Unit 5: Sports & Nutrition | Tag: Very High***

Q14. In the administration of the Harvard Step Test, what is the standard prescribed height of the stepping bench for women? (A) 20 inches (50 cm) (B) 16 inches (40 cm) (C) 18 inches (45 cm) (D) 14 inches (35 cm) *Unit 6: Test and Measurement in Sports | Tag: Very High***

Q15. The total volume of air inhaled or exhaled by the lungs per minute is scientifically referred to as: (A) Vital Capacity (B) Tidal Volume (C) Minute Volume (D) Residual Volume *Unit 7: Physiology & Injuries in Sport | Tag: High***

Q16. Assertion (A): Bending the knees while landing from a high jump significantly reduces the impact force on the legs. **Reason (R):** Bending the knees increases the time of impact, thereby reducing the rate of change of momentum according to Newton's Second Law of Motion. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 8: Biomechanics and Sports | Tag: Very High***

Q17. Assertion (A): Roughage (dietary fiber) does not provide any caloric energy to the body but is still considered an essential component of a balanced diet. **Reason (R):** Roughage adds bulk to the food, helps in the smooth excretion of waste, and effectively prevents constipation. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 5: Sports & Nutrition | Tag: Very High***

Q18. Assertion (A): Fast-twitch muscle fibers contract extremely quickly and produce explosive power, but they fatigue very rapidly. **Reason (R):** Fast-twitch muscle fibers rely primarily on aerobic metabolism (oxygen) for continuous and sustained energy production. (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true, but (R) is not the correct explanation of (A). (C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true. *Unit 7: Physiology & Injuries in Sport | Tag: High***

SECTION B: Very Short Answer Type (10 Marks: Attempt any 5 out of 6 x 2)

Word limit: 60-90 words.

Q19. Define 'Combination Tournaments'. List any four specific types of combination tournaments used in sports. *Unit 1: Management of Sporting Events | Tag: High***

Q20. Enlist any two major psychological benefits that women achieve by actively participating in competitive sports. *Unit 2: Children & Women in Sports | Tag: Very High***

Q21. State any two contraindications (medical conditions where it should be avoided) for performing *Tadasana* (Palm Tree Pose). *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | Tag: High***

Q22. Briefly explain the specific role of a 'Speech Therapist' in assisting Children with Special Needs (CWSN). *Unit 4: Physical Education & Sports for CWSN | Tag: High***

Q23. What is the primary purpose of the 'Johnson-Methney Test of Motor Educability'? List any two of its test items. *Unit 6: Test and Measurement in Sports | Tag: Medium***

Q24. Clearly differentiate between 'Vital Capacity' and 'Minute Volume' in relation to the human respiratory system. *Unit 7: Physiology & Injuries in Sport | Tag: Very High***

SECTION C: Short Answer Type (15 Marks: Attempt any 5 out of 6 x 3)

Word limit: 100-150 words.

Q25. Explain the 'Staircase Method' of drawing fixtures in a League tournament. What is its main mathematical advantage? *Unit 1: Management of Sporting Events | Tag: Very High***

Q26. Discuss 'Osteoporosis' in the context of the Female Athlete Triad. What are its primary physiological causes? *Unit 2: Children & Women in Sports | Tag: Very High***

Q27. Clearly differentiate between Macro-nutrients and Micro-nutrients by providing two points of difference and one specific example of each. *Unit 5: Sports & Nutrition | Tag: Very High***

Q28. Detail the procedure and the specific scoring method of the 'Push-up' test designed for boys under the SAI Khelo India Fitness Test battery. *Unit 6: Test and Measurement in Sports | Tag: High***

Q29. Differentiate between an 'Impacted Fracture' and an 'Oblique Fracture'. Give brief descriptions of how both injuries typically occur on the field. *Unit 7: Physiology & Injuries in Sport | Tag: Medium***

Q30. Define 'Centre of Gravity' (CG). Explain practically how the position of the centre of gravity affects the dynamic stability of a wrestler during a combat bout. *Unit 8: Biomechanics and Sports | Tag: Very High***

SECTION D: Case-Study Based Questions (12 Marks: 3 x 4)

All 3 questions are compulsory. Internal choice is provided in one sub-part of each.

Q31. A reputed sports academy is hosting a massive Zonal Hockey Tournament using a "Knock-out cum League" combination format. There are 16 teams participating in total. The organizing committee divides these 16 teams equally into 4 separate zones (Zones A, B, C, and D) containing 4 teams each. **(i)** During the initial knock-out phase within a single zone (which has 4 teams), how many matches will be played to declare the zonal winner? **(ii)** After the initial knock-out phase is over, the 4 respective zonal winners will compete against each other in a single league format. Calculate the total number of matches that will be played in this final league phase. **(iii)** Name the specific organizing committee responsible for registering these teams and preparing the tournament brackets. **(iv)** What will be the total number of matches played in the entire tournament (combining all zones and the final league)? **OR** Write the exact formula for calculating the total number of matches in a pure single knock-out tournament. *Unit 1: Management of Sporting Events | Tag: Very High***

Q32. Mr. Sharma, a 45-year-old corporate employee, visits an orthopaedic doctor complaining of severe stiffness in his lower spine and sharp, shooting pain radiating down his leg (Sciatica/Back pain). Due to prolonged sitting at a desk, his back muscles have weakened. The doctor recommends a combination of physiotherapy and practicing specific yoga asanas to decompress his spine and restore flexibility. **(i)** Name any two yoga asanas specifically prescribed in the syllabus for alleviating Back Pain. **(ii)** Identify one specific lying-down (prone) asana that is highly effective in strengthening the back and spinal column. **(iii)** If Mr. Sharma was suffering from poor insulin production instead of back pain, which lifestyle disease would he be diagnosed with? **(iv)** Mention the primary health benefit of performing *Bhadrasana*. **OR** Name an asana recommended for back pain that involves twisting the spine while in a seated position. *Unit 3: Yoga as Preventive Measure for Lifestyle Disease | Tag: Very High***

Q33. A high school physics teacher is explaining the concepts of sports biomechanics to her class using a live football match. She points out that after a player kicks the football, it rolls along the grass but eventually comes to a complete halt due to invisible opposing forces. She also explains that the football players intentionally wear boots with studs (cleats) on the soles. **(i)** The opposing resistance force that eventually stops the rolling football on the grass is called _. ***(ii)** Football players wear studded cleats purposefully to increase which specific type of friction between their shoes and the ground? **(iii)** The parabolic flight path of the football when kicked high into the air toward the goalpost is scientifically known as its _.* **(iv)** Which of Newton's Laws states that the football will remain stationary on the penalty spot until a player applies muscular force to kick it? **OR** Define dynamic friction. *Unit 8: Biomechanics and Sports | Tag: High***

SECTION E: Long Answer Type (15 Marks: Attempt any 3 out of 4 x 5)

Word limit: 200-300 words.

Q34. The spinal column is prone to various misalignments due to poor lifestyle habits. Elaborate on the common postural deformities related strictly to the spine: Kyphosis, Lordosis, and Scoliosis. Detail their visible anatomical symptoms and suggest at least two corrective physical exercises or yoga asanas for each deformity. *Unit 2: Children & Women in Sports | Tag: Very High***

Q35. What is the significance of a Balanced Diet in an athlete's life? Elaborate comprehensively on the 'Pitfalls of Dieting' (such as extreme caloric restriction) and the physiological dangers of 'Food Intolerance', explaining exactly how they negatively impact an athlete's health and sports performance. *Unit 5: Sports & Nutrition | Tag: High***

Q36. The Rikli and Jones Senior Citizen Fitness Test is a benchmark battery for assessing the functional fitness of elderly adults. Describe the complete test battery, detailing the exact physical component measured by each of the six test items (e.g., Chair Stand Test, Arm Curl Test, Chair Sit & Reach, Back Scratch, 8-Foot Up & Go, and 6-Minute Walk Test). *Unit 6: Test and Measurement in Sports | Tag: Very High***

Q37. What are 'Physiological Factors' in sports? Discuss comprehensively any four distinct physiological factors (e.g., muscle fiber type, joint structure, internal organ capacity, ATP reserves) that determine the 'Flexibility' and 'Speed' of an elite athlete. *Unit 7: Physiology & Injuries in Sport | Tag: Very High***

Answer Key

Q1. (B) Knock-out cum League **Q2.** (C) Bow Legs (Genu Varum) **Q3.** (B) Bhujangasana **Q4.** (C) Eunice Kennedy Shriver **Q5.** (B) Sodium **Q6.** (C) Specially constructed wooden testing box **Q7.** (A) Transverse Fracture **Q8.** (D) Fluid friction **Q9.** (B) Announcement Committee **Q10.** (B) Osteoporosis **Q11.** (C) It is a Pranayama (breathing technique) for purifying the energy channels. **Q12.** (D) FIFA **Q13.** (D) Vitamin K **Q14.** (B) 16 inches (40 cm) **Q15.** (C) Minute Volume **Q16.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q17.** (A) Both (A) and (R) are true and (R) is the correct explanation of (A). **Q18.** (C) (A) is true, but (R) is false. (Reason: Fast-twitch fibers rely on *anaerobic* metabolism, without oxygen, which is why they fatigue quickly).

Q19. Combination Tournaments are conducted when matches are to be played on a group or zonal basis, merging two tournament formats. **Types:** 1. Knock-out cum Knock-out. 2. League cum League. 3. Knock-out cum League. 4. League cum Knock-out. **Q20.** 1. **Boosts Self-Esteem:** Mastering physical skills instils a high sense of confidence and positive body image. 2. **Stress Reduction:** Regular physical exertion releases endorphins, alleviating symptoms of depression, stress, and anxiety. **Q21.** 1. Individuals suffering from low blood pressure or experiencing frequent headaches/migraines should avoid it. 2. Individuals suffering from severe insomnia should avoid performing Tadasana. **Q22.** A **Speech Therapist** evaluates and treats CWSN who suffer from communication disorders, speech impediments (like stuttering), or swallowing difficulties. They help the child articulate words better, boosting their confidence to communicate effectively with teammates and coaches on the field. **Q23. Purpose:** To measure the fundamental motor educability (ability to learn new motor skills easily) of an individual. **Items:** 1. Front Roll. 2. Back Roll. (Others include Jumping Half-Turn, Jumping Full-Turn). **Q24. Vital Capacity:** The maximum volume of air a person can exhale forcefully after a maximal inhalation. **Minute Volume:** The total volume of air inhaled or exhaled by the lungs in exactly one minute (Tidal Volume $\times$ Respiratory Rate).

Q25. Staircase Method: The fixtures are made in a step-by-step ladder/staircase formation. Team 1 plays with all other teams in the first step. Then Team 2 plays with the remaining teams in the next step down, and so on. **Advantage:** It is the easiest mathematical method because no 'Byes' are calculated or given, regardless of whether the total number of teams is odd or even. **Q26. Osteoporosis** in the Female Athlete Triad is a severe weakening of the bones, making them porous, fragile, and highly prone to stress fractures. **Causes:** It is primarily caused by low estrogen levels (a direct result of amenorrhea/lack of menstruation) combined with poor dietary intake of calcium and Vitamin D due to eating disorders. **Q27. Macro-nutrients:** Required by the body in large amounts; they provide bulk energy. Example: Carbohydrates, Proteins, Fats. **Micro-nutrients:** Required by the body in very small/trace amounts; they regulate bodily functions but provide zero energy. Example: Vitamins (A, C, D) and Minerals (Iron, Calcium). **Q28. Procedure:** The boy takes a front-leaning plank position with hands shoulder-width apart, arms straight, and toes tucked. The body must remain perfectly straight. He lowers the body until the chest is approx 2 inches from the floor, then pushes back up. **Scoring:** The total number of correctly performed, continuous push-ups executed without breaking form or resting is recorded. **Q29. Impacted Fracture:** The broken ends of a bone are forcefully jammed or driven into each other. Occurs due to a heavy, straight-axis impact (e.g., landing stiff-legged from a high jump). **Oblique Fracture:** A diagonal break across the bone shaft. Occurs from a severe twisting force or angled trauma during a tackle. **Q30. Centre of Gravity (CG)** is an imaginary point in a body where its entire weight is evenly balanced. **Application:** To maintain stability, a wrestler widely spreads their legs (increasing the base of support) and deeply bends their knees (lowering the CG). The lower the CG and wider the base, the harder it is for the opponent to topple them.

Q31. (i) Matches in one zone = $N - 1 = 4 - 1 = 3$ matches. (ii) League matches for 4 winners = $N(N-1)/2 = 4(3)/2 = 6$ matches. (iii) Registration & Fixture Committee (or Technical Committee). (iv) Total Matches = (4 zones $\times$ 3 knock-out matches) + 6 league matches = $12 + 6 = 18$ matches. **OR** Formula: $N - 1$ (where N is total number of teams). **Q32.** (i) Tadasana, Ardh Chakrasana, Bhujangasana, Vakrasana (Any two). (ii) Bhujangasana (Cobra Pose) or Makarasana (Crocodile Pose). (iii) Diabetes. (iv) Bhadrasana strengthens the pelvic organs and relieves stiffness in the knees and hip joints, easing back tension. **OR** Vakrasana / Ardha Matsyendrasana. **Q33.** (i) Rolling friction. (ii) Static friction (or simply grip/sliding friction resistance). (iii) Trajectory (or Parabola). (iv) Newton's First Law (Law of Inertia). **OR** Dynamic friction is the resistance that acts between two surfaces when they are in active relative motion (sliding or rolling) against each other.

Q34. 1. Kyphosis (Round upper back/Hunchback): An exaggerated posterior outward curvature of the thoracic spine. Shoulders slump forward. *Corrective Exercises:* Practicing Chakrasana (Wheel pose), Dhanurasana (Bow pose), or swimming to open the chest and pull the shoulders back. **2. Lordosis (Hollow back):** An exaggerated inward curvature of the lumbar spine (lower back), causing the abdomen to protrude forward. *Corrective Exercises:* Practicing Halasana (Plough pose), Paschimottanasana, or performing regular sit-ups to strengthen the abdominal core and stretch the lower back. **3. Scoliosis:** An abnormal lateral (sideways) "S" or "C" shaped curvature of the spine. One shoulder or hip may appear higher than the other. *Corrective Exercises:* Practicing Ardh Chakrasana (bending to the opposite side of the curve), Trikonasana, or hanging from a horizontal bar to decompress and straighten the spinal column. **Q35.** A **Balanced Diet** contains correct proportions of carbohydrates, proteins, fats, vitamins, minerals, and water, crucial for fueling performance, tissue repair, and immune function. **Pitfalls of Dieting:** Athletes sometimes adopt crash diets (extreme caloric restriction, skipping meals, cutting out carbs) for rapid weight loss. This drastically lowers the Basal Metabolic Rate (BMR), depletes glycogen stores, causes severe dehydration, and leads to muscle catabolism (muscle loss), instantly ruining endurance and explosive strength. **Food Intolerance:** It is the digestive system's inability to process certain food compounds (e.g., lactose in milk or gluten in wheat) due to missing enzymes. If an athlete consumes such foods, it causes immediate stomach cramps, nausea, bloating, and diarrhea. This gastrointestinal distress destroys an athlete's focus, hydration levels, and physical capacity during a competitive event. **Q36.** The Rikli and Jones Senior Citizen Fitness Test contains 6 specific items to assess the functional daily capacity of older adults (60+ years): **1. Chair Stand Test:** Evaluates *lower body strength* (essential for getting out of a chair/bus). The participant does maximum full sit-to-stand movements in 30 seconds. **2. Arm Curl Test:** Evaluates *upper body strength*. Men curl an 8 lb dumbbell, women a 5 lb dumbbell, doing maximum bicep curls in 30 seconds. **3. Chair Sit and Reach Test:** Evaluates *lower body flexibility* (hamstrings). Participant sits on a chair with one leg extended and reaches for the toes. **4. Back Scratch Test:** Evaluates *upper body (shoulder) flexibility*. The participant tries to touch the fingers of both hands together behind the back. **5. 8-Foot Up and Go Test:** Evaluates *agility and dynamic balance*. Participant stands from a chair, walks 8 feet around a cone, and sits back down as fast as possible. **6. 6-Minute Walk Test:** Evaluates *aerobic/cardiovascular endurance*. Participant walks as far as possible along a marked course in exactly 6 minutes. **Q37. Physiological Factors** are the internal organic and functional capacities of the body systems that dictate physical performance. **Factors Determining Flexibility:** 1. **Joint Structure:** The anatomical structure of a joint limits its range. Ball-and-socket joints (shoulder) inherently have greater flexibility than hinge joints (knee). 2. **Age and Gender:** Flexibility naturally decreases with age due to tendon stiffness. Females are generally more flexible than males due to anatomical pelvic differences and hormones. **Factors Determining Speed:** 3. **Muscle Fiber Composition:** Speed is strictly determined by the genetic ratio of Fast-Twitch (White) muscle fibers. A higher percentage of fast-twitch fibers allows for explosive, rapid muscular contractions essential for sprinting. 4. **Neuromuscular Coordination:** The central nervous system's ability to send rapid impulses to muscles (motor unit recruitment) dictates the reaction time and contraction speed of the athlete. Efficient coordination eliminates internal muscle resistance, allowing faster movement.